

Financing the transition

Nine Japanese cases in industrial decarbonisation, and what they reveal about who owns, who pays and who carries the risk — with questions for the City of Malmö

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Meeting Malmö, 9 September 2026

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Evidence base A study visit to Kitakyushu on 22 September 2023 by the Public Finance Study Group, Kyoto University, together with the city and company documents listed in Part V; and the author's own research on Kitakyushu's industrial structure

Context Nordic field study, 7–11 September 2026 — Danish Energy Agency · Copenhagen Business School · Malmö · Lolland

§ Executive summary

Kitakyushu is Japan's clearest example of a heavy-industry city that rebuilt its identity around the environment. It is also, on the economic evidence, a warning. This report sets out nine cases drawn from the city and its region, examines each for its *ownership and financing structure* rather than its targets, and arrives at a single argument I would like to test against Malmö's experience.

The argument. Japanese municipalities have weak fiscal autonomy and, since electricity liberalisation, nothing resembling a Nordic municipal utility. Kitakyushu has therefore solved decarbonisation by *buying services rather than owning assets* — third-party ownership of solar, batteries and even air conditioning on public buildings, under a rule that the resulting tariff must not exceed the electricity price the city already pays. It works: the rollout is fast, the municipal balance sheet is untouched, and 626 public buildings across eighteen municipalities were on renewable electricity by mid-2023. But the city ends up with contracts rather than capital, and the pattern repeats one level up: the region's flagship offshore wind farm is owned by a consortium of five national utilities and contractors, in which the municipality's role was to run the tender and provide the port.

The counter-evidence. Two cases cut against that reading. The reuse-panel demonstration at the Hibikinada Biotope shows the same third-party model being extended to second-life assets, with a leasing company inside the structure — the beginnings of a financing industry rather than a procurement trick. And Gotō, a small island city in Nagasaki, has produced something Kitakyushu has not: a locally-owned maintenance firm, a citizens' electricity retailer, and a training association that deliberately creates competitors. Value is retained locally there in a way that 940,000-person Kitakyushu has never achieved.

The long record. Kitakyushu has run an environment-and-knowledge-led growth strategy for three decades through its Science and Research Park and the agency FAIS. Gross product has nonetheless grown slowly; what actually expanded was health care, real estate and public services, not the targeted high-value manufacturing. My tentative explanation is that the programme's success criterion was commercialisation rather than value added, that no private capital layer ever formed, and that the business services a technology cluster needs never agglomerated — so the city substituted for them itself, and thereby capped the scale at what a shrinking municipal budget can fund.

What I want from Malmö. Malmö took the opposite route on almost every one of these choices. Part IV puts fourteen questions, grouped by theme. The single one I care most about: does a city that keeps ownership end up with more capability than a city that buys services — or only with more debt?

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Framing

Why this comparison, what Kitakyushu is, and how it got here.

1 The question

Malmö and Kitakyushu are not obvious twins, but their problem was the same. Each was a second city built on a single industry — shipbuilding in Malmö, integrated steelmaking in Kitakyushu — and each lost the certainty that industry provided. Both answered by redefining themselves as environmental cities. Both are now judged on whether that redefinition produced an economy or only a reputation.

I have studied Kitakyushu since my undergraduate work, and my interest is deliberately narrow. I am not asking what the city declares; declarations are cheap and Kitakyushu has made many good ones. I am asking three questions of every project: **who owns the asset, who supplies the capital, and who is left holding the risk**. Those three answers determine whether a transition builds durable municipal capability or merely rents it.

The reason to ask them of Japan specifically is that the Japanese municipal toolkit is unusually thin. Local governments have limited own-source revenue and limited borrowing discretion. Since electricity liberalisation there is no municipal utility in the Nordic sense — Kitakyushu holds a stake in a local retailer, Kitakyushu Power, but not in generation, networks or heat. What a Japanese city has to work with is its buildings, its land, its port, its convening power, and its procurement. Everything in Part II is a variation on making those five things do the work that a balance sheet does elsewhere.

2 Kitakyushu at a glance

Established	1963, by the merger of five cities of equal standing (Moji, Kokura, Wakamatsu, Yahata, Tobata) — an unusual origin that left the city polycentric and without a single dominant centre
Area	491.71 km²
Population	c. 940,000 (January 2022); c. 490,000 households. A designated city in long-term decline from a peak above 1.06 million in 1979
Principal industries	Iron and steel, chemicals, cement, ceramics, automotive
Greenhouse gas emissions	14.78 Mt CO ₂ e (2019)
Industrial share of emissions	64% — against 32% for Japan as a whole (national total 1.21 Gt CO ₂ e, 2019)
Headline target	–47% GHG by FY2030 against FY2013; net zero by 2050
Installed renewables (city)	Wind 24,852 kW and solar 255,757 kW at end-March 2017; target 1,302–1,402 MW by FY2030

That 64% is the most important number in this report, and it recurs in every case that follows. Two-thirds of Kitakyushu's emissions sit inside a handful of private industrial installations that the city does not own, cannot regulate directly, and cannot afford to lose. Every instrument in Part II is shaped by that fact: the city works intensively on the

third of emissions it can reach, and tries to influence the rest through industrial policy rather than climate policy. When Kitakyushu talks about decarbonisation it is, in a precise sense, talking about industrial strategy.

3 Four phases

PERIOD	PHASE	WHAT HAPPENED
1901–1963	Building the steel city	The state-owned Yawata Steel Works opens in 1901 and becomes the core of Japanese industrialisation. Five towns grow together around it and merge in 1963.
1960s–1970s	Pollution, and the citizens' response	Dokai Bay is declared a "sea of death"; the sky over Yahata is described as having seven colours. Local women's associations document the damage and force the issue onto the political agenda, producing pollution-control agreements between the city and industry ahead of national law. The environmental identity was imposed from below, not chosen.
1980s–2000s	Environment as an industry	The clean-up is converted into an export. The Eco-Town project (from 1997) turns the Hibikinada waterfront into a recycling industrial estate; the Science and Research Park and its agency FAIS attempt a knowledge economy; the city becomes a hub for environmental cooperation with Asian cities. Eco-Model City (2008), SDGs Future City (2018).
2010s–	Energy as an industry	Green Energy Port Hibiki (from 2011) reorients the port towards offshore wind. Zero Carbon City declaration (October 2020); climate emergency declaration (June 2021); revised climate plan (August 2021); Green Growth Strategy (February 2022); selection as a national <i>Decarbonisation Leading Area</i> (April 2022).

THE NATIONAL POLICY BACKDROP, COMPRESSED

The Japanese offshore wind sector is very young, and the Kitakyushu cases only make sense against it: feed-in tariff introduced 2012; Port and Harbour Act amended 2016 to allow competitive tendering of port water areas; 5th Basic Energy Plan 2018 (wind at 1.7%); the *Renewable Sea Area Utilisation Act* in force 2019 — widely called Japan's "offshore wind year one"; July 2020, a policy to retire around 90% of inefficient coal capacity; September 2020, four base ports designated (Kitakyushu, Akita, Noshiro, Kashima); December 2020, a national target of up to 45 GW of offshore wind by 2040; 2020 Port Act amendment extending port occupancy rights from 20 to 30 years. Kitakyushu began its port programme in 2011, ahead of all of it.

PART II

Nine cases

Each case is read the same way: what it is, the numbers, the ownership and financing structure, what it demonstrates, and what

remains unresolved.

CASE 1 · THE MUNICIPAL ESTATE

Third-party ownership at scale

Decarbonisation Leading Area and the "100% Renewable Kitakyushu Model" — how a city with no capital decarbonises 3,600 buildings.

WHAT IT IS

In April 2022 Kitakyushu was selected in the first cohort of the national *Decarbonisation Leading Area* programme, which requires net-zero electricity in the residential and commercial sector by FY2030 and is intended to produce a replicable model — the government's phrase is a "decarbonisation domino". Kitakyushu applied jointly with **17 neighbouring municipalities**, eighteen in total.

Scope approximately **3,600 public facilities** across the eighteen municipalities, plus the recycling companies of Eco-Town · **New renewable capacity** 37 MW · **PV sites** c. 290 in Kitakyushu, c. 130 in the 17 neighbours · **Renewable electricity share** 100% for the target estate · **Funding** the national Regional Decarbonisation Transition and Renewable Energy Promotion Grant

THE STRUCTURE

The city does not buy solar panels. It buys electricity from panels someone else owns that happen to sit on its roofs. Its own slogan is a shift "**from ownership to use**".

- Equipment company** owns and installs the PV, battery or air-conditioning unit
 - **JV / consortium** operates and manages it, with IoT monitoring
 - **Retailer** (Kitakyushu Power and others) holds the electricity supply contract
 - **Public or private facility** pays an electricity tariff only — no capital expenditure

Above this sits a three-step ladder that any building can be moved along. **Step 1** is 100% renewable electricity supplied from generation *inside the city*, including waste-to-energy plants, through Kitakyushu Power. **Step 2** adds PV plus a battery under third-party ownership, charging when power is cheap and discharging when it is expensive. **Step 3** adds energy-efficient equipment, also third-party owned, reducing total consumption and therefore the volume of renewable electricity that must be procured.

RESULTS TO MID-2023

INDICATOR	DATE	RESULT
Public facilities on 100% renewable electricity (18-municipality area)	1 Jul 2023	626 facilities (586 in Kitakyushu itself)
CO ₂ avoided by those facilities	1 Jul 2023	10,936.5 t/year
Firms certified under the Decarbonised Electricity Certification scheme	1 May 2023	36 companies, 48 sites ; c. 44,550 t CO ₂ /yr
PV installed under the Leading Area programme	FY2022	16 public facilities (350 kW); 1 Eco-Town firm (200 kW)
Batteries installed	FY2022	14 public facilities (230 kWh); 1 private site (10 kWh)
Air conditioning installed	FY2022	28 public facilities; grant spend ¥100m and ¥370m

THREE DESIGN CHOICES THAT MATTER

- **A hard price rule.** The city will only sign a PPA where the resulting tariff is at or below the electricity price it already pays. Decarbonisation is not permitted to cost more. This is what has kept the programme politically durable — and also what limits how fast it can spread, since it excludes every building where the arithmetic does not work.
- **Speed, not just cost.** The clearest case is air conditioning for school kitchens, installed for heat-stress protection of staff. Conventional procurement would have taken about 13 years at roughly 10 schools a year, at c. ¥48m/year. Under third-party ownership the rollout takes about 2 years at c. ¥44m/year, and IoT monitoring extends equipment life from 13 to 15 years. The saving is modest; the eleven years are the point.
- **Scale by aggregation.** A city of 940,000 cannot make a low-cost PPA model bankable on its own estate. Bundling eighteen municipalities and roughly 3,600 buildings creates a portfolio large enough for private counterparties to bid on. *This is the single most transferable idea in the report.*

A WORKED EXAMPLE — HIAGARI CITIZENS' CENTRE (STEP 2)

12 kW of PV and one 16.4 kWh stationary battery on a building consuming about 40,000 kWh a year. Roughly 13,000 kWh a year is self-consumed, giving about **33% electricity self-sufficiency**. The building is also a designated evacuation shelter: on a full charge the battery can run office lighting for 12 hours and charge the mobile phones of 50 evacuees for 18 hours. In Japan, resilience is frequently the argument that actually unlocks the budget, with carbon as the secondary benefit.

THE CERTIFICATION SCHEME — RECOGNITION CONVERTED INTO MONEY

A company on a decarbonised electricity tariff receives a sticker, a logo and publication on the city's channels. It also receives eligibility for a municipal finance facility and additional points in the scoring of four separate municipal grant and business-selection schemes. The city spends almost nothing and converts reputation into access to money it already controls. The first hundred firms are designated "decarbonisation pioneer companies".

UNRESOLVED

A long service contract sits awkwardly with annual municipal budgeting; the city ends the period with contracts rather than assets; and the model reaches only buildings where the price rule holds. Nobody has yet shown what happens at contract expiry.

The reuse-panel demonstration at Hibikinada Biotope

Third-party ownership extended to used equipment — and a leasing company brought inside the structure.

WHAT IT IS

A small demonstration beside the nature centre at the Hibikinada Biotope: a **5 kW array of reused solar panels**, roughly ten years into their service life, generating for on-site self-consumption. The site is itself one of the facilities covered by the 100% Renewable Kitakyushu Model.

WHY IT EXISTS

Japan's first wave of feed-in-tariff solar is now reaching the end of its 20-year contracts, and a very large volume of panels will come off roofs and fields over the next decade. The demonstration has three stated purposes: to find productive uses for megasolar assets after FIT expiry; to build and stress-test a *third-party ownership model for reused panels*; and to work out the rules such a market would need. It measures two things — the degradation and output efficiency of decade-old panels, and their cost.

THE STRUCTURE — AND WHY IT IS THE MOST INTERESTING LINE IN THIS REPORT

- Shinryo** (Eco-Town recycler, Mitsubishi Chemical group) supplies and processes the panels
- **Tokyo Century** (leasing and finance company) — the asset owner and financier
- **Kitakyushu Power** (local retailer) — the supply relationship
- **City of Kitakyushu** — host site, convenor, and the party defining the public purpose

Case 1 is a procurement technique. This is the same technique with a *financial institution inside it*. A leasing company taking ownership of second-life assets is the beginning of an asset class: if used panels can be underwritten, they can be financed, and if they can be financed the circular-economy claims in Case 3 acquire a balance sheet. Kitakyushu's weakness across thirty years, as Case 9 argues, has been precisely the absence of private capital. This is the smallest case in the report and possibly the most consequential.

UNRESOLVED

5 kW is a demonstration, not a market. Whether reused panels can be insured, warranted and depreciated on terms a lessor will accept at scale is exactly what the experiment is meant to find out, and it had not reported when these materials were produced.

Eco-Town: Shinryo and Recycle Tech

The clean-up converted into an industry — and the industry now recycling the energy transition's own waste.

THE FIRMS

Shinryo Co., Ltd.

Founded October 1964. Head office in Yahatanishi-ku, Kitakyushu. Capital ¥500m, wholly owned by Mitsubishi Chemical. Sales ¥25.9bn (FY2021), c. 1,254 staff (March 2022). Neutralises and converts waste acid, alkali and oil into fuel and cement feedstock; precision cleaning of semiconductor and display equipment parts; wafer reclamation; fine chemicals.

Recycle Tech Co., Ltd.

Founded December 1998; capital ¥30m; owned by Shinryo and Ricoh. Operating in the Hibikinada Eco-Town since 1998 under the national Eco-Town designation. Annual capacity 5,400 t of office equipment — c. 3,600 copiers and 4,500 PCs and printers per month — dismantled by hand to reach **99% resource recovery**. Employment of people with disabilities is an explicit part of the operating concept.

SOLAR PANEL RECYCLING

The newer line is photovoltaic panel recycling, planned at **90,000 panels a year**, with a stated recycling rate above **99%**. Panels are sorted into reuse (sold into the second-hand market) and recycling; the high-grade line separates the aluminium frame and sheet glass, thermally decomposes the EVA encapsulant, and recovers silver, copper, glass cullet and silicon cells.

Per 1 MW of decommissioned panels (c. 86 t): aluminium **c. 13 t**, glass **c. 54 t**, silver and copper **c. 1 t** · recycling rather than landfill or road-base use avoids **c. 200 t CO₂** · the same 1 MW of megasolar avoids **c. 18,000 t CO₂** over 20 years against coal generation

WHAT IT DEMONSTRATES

Eco-Town is the one part of Kitakyushu's strategy that indisputably produced an industry, and it did so by the least fashionable route: a national designation, a designated site, and large incumbent chemical and electronics firms willing to put subsidiaries there. It is not a start-up story. It is also now closing a loop with Case 1 — the panels the city is installing under third-party ownership will be handled at end of life by firms three kilometres away, and second-life panels and EV batteries from those firms are being cascaded back into the programme.

UNRESOLVED

The recovered materials are commodities with thin margins, and the volumes depend entirely on a decommissioning wave whose timing is set by national FIT policy, not by the city.

Kitakyushu Hibikinada Offshore Wind Farm

The largest offshore wind farm in Japan at completion — tendered by the municipality, owned by five national companies.

25 turbines of 9,600 kW, maximum output **220,000 kW** · sea area c. **2,700 ha**, 1–10 km north–south by 11 km east–west, water depth 8–30 m · rotor diameter **174 m**, blade tip c. **200 m** above sea level · jacket (fixed-bottom) foundations · output c. **500 GWh/year** ≈ **170,000 households** ≈ **40% of the city's households** · **20-year** generation business · construction began **13 March 2023**, commissioning within FY2025

HOW IT CAME ABOUT

Not through the national auction system, which did not yet exist. In **February 2017** the *City of Kitakyushu* ran a public tender for the port water area — possible because of the 2016 Port and Harbour Act amendment — and selected the consortium that became Hibiki Wind Energy K.K., incorporated in April 2017. The company then carried out wind and marine surveys and environmental impact assessment, obtained wind farm certification, and started construction in March 2023. Six years from tender to first steel.

WHO OWNS IT

SHAREHOLDER	SHARE	WHAT THEY ARE
J-POWER (Electric Power Development Co.)	40%	National wholesale generator
Kyuden Mirai Energy	30%	Renewables arm of Kyushu Electric Power
Hokutaku	10%	Wind turbine O&M specialist
Saibu Gas	10%	Regional gas utility
Kyudenko	10%	Electrical engineering contractor

The construction chain is similarly national: Vestas Japan supplies the turbines; a Penta-Ocean / Nippon Steel Engineering joint venture builds and installs the foundations; J-POWER Hitec does the onshore electrical work; a Penta-Ocean / Wakachiku joint venture builds the O&M port; Tokyo Kisen operates the crew transfer vessels.

THE MUNICIPALITY'S ACTUAL ROLE

Kitakyushu's contribution was the water area, the tender, the base port, and legitimacy — the fisheries negotiations, the citizen briefings, the local-firm matching sessions with manufacturers, the seminars and the links to local education. The company describes its ambition as a wind farm "that walks with the region". **The city holds no equity.** It captured the project by owning the seabed access and the port, not by investing.

THE COMPARISON I WOULD LIKE TO DRAW

This is, I think, the sharpest single contrast with Denmark and Sweden. A Japanese municipality can convene a 220 MW project, shape it, and host it — and end up with property tax, some local contracts and a photograph of the mayor at the groundbreaking. Whether that is a failure or simply a different and defensible division of labour is a question I cannot answer from the Japanese side alone. Note also the capacity arithmetic: 25 × 9.6 MW is 240 MW of turbines declared at a maximum output of 220 MW — the grid connection, not the wind, sets the ceiling.

Green Energy Port Hibiki

A port used as the instrument a municipality actually possesses.

WHAT IT IS

A programme begun by the city in **2011** — before the feed-in tariff, eight years before Japan's offshore wind legislation — to convert the Hibikinada district into a comprehensive base for the wind industry, built on four functions: import/export and domestic transshipment; turbine marshalling and load-out; operations and maintenance; and manufacturing of turbines, components and structures.

In **September 2020** Kitakyushu Port was designated a base port under the Port and Harbour Act — one of four nationally (with Akita, Noshiro and Kashima) and **the only one in western Japan**. Designation matters physically as well as legally: turbine components now exceed 60 m in length and 400 t in weight, and require reinforced quays and large back-up storage yards that only a designated port will receive public investment to build.

PHASES

Phase 1 (2013–)	A demonstration zone: onshore testing of offshore turbines, an O&M base, training facilities, and a solar–wind hybrid demonstration, procured by inviting proposals that would lead to industrial agglomeration.
Phase 2 (2016–)	Attracting Japan's first full-scale offshore wind farm into the port area — Case 4 — explicitly to create domestic demand and give the agglomeration something to serve.
Phase 3	Building the base port and the surrounding industrial land, and recruiting firms and vessels to it.
Phase X	Adapting to floating foundations and to turbines in the 12–16 MW class, whose blade tips exceed 300 m.

TURBINE SCALE AS A PROXY FOR THE WHOLE STORY

Ten 1.5 MW GE turbines installed by NS Wind Power Hibiki in March 2003 reached 100 m at the tip. Two 3.3 MW Vestas machines installed by Wind Energy Research Park in January 2018 reached 140 m. The 9.5 MW class planned for the wind farm reaches 200 m. The 12–16 MW machines now in development exceed 300 m. Every one of those steps re-specifies the quay, the vessels and the factories — which is why a port strategy has to be a rolling one.

THE SUPPLY CHAIN THE PORT ASSEMBLED

Nippon Steel Engineering (jacket foundations, Wakamatsu); thyssenkrupp rothe erde Japan (large slewing bearings, Hibikinada); Furukawa Electric Industrial Cable (tower and nacelle cables, Moji); Yaskawa Electric (generators and electrical products, Yukuhashi); Regency Steel Japan (steel structures, Tobata); Ishibashi Manufacturing (gearboxes, Nōgata); Penta-Ocean (SEP jack-up vessel); Tokyo Kisen (crew transfer vessels); Hokutaku (O&M). Distances from Hibikinada to the other base ports — Noshiro 630, Kashima 620, Akita 580 nautical miles — and to Nagasaki 155 and Taipei 750, define the market it can actually serve.

WHAT IT DEMONSTRATES, AND WHAT IS UNRESOLVED

This is the clearest case of a Japanese municipality using an asset it genuinely owns to shape private investment, and doing so a decade ahead of national policy. Whether it pays depends on something Kitakyushu does not control: how much offshore wind Japan actually builds in the western sea areas, and whether floating foundations shift the industrial centre of gravity elsewhere.

FLOAT RAISER — a state-subsidised construction vessel

How Japan builds an industrial capability when no private party will fund the first unit.

WHAT IT IS

A semi-submersible spud barge built from FY2016 with a subsidy from the Ministry of the Environment's programme for cost reduction and diffusion of low-carbon floating offshore wind. Owned and operated by Offshore Wind Farm Construction Co., Ltd., a vehicle of Toda Construction and Yoshida-gumi. The name means the vessel (*raiser*) that stands a float upright.

110.0 m × 43.0 m, depth 6.8 m, draught 4.7 m, gross tonnage **12,300 t** · deck area c. **3,890 m²**, deadweight c. **13,500 t** · roll-on capability for loads up to c. **5,000 t**, from either the long or the short side · submerges to **7.4 m** below deck level, and can do so trimmed fore or aft · four 40 m spuds; ballast pumping 1.2–6.7 m/h

WHAT IT DOES

A hybrid spar float for a floating wind turbine is built on land, rolled onto the barge on multi-axle trailers with the ballast adjusted as it comes aboard, towed offshore, and then floated off as the barge submerges. The same vessel can launch caissons, jackets and work vessels, so it serves fixed-bottom construction and general marine works as well.

WHAT IT DEMONSTRATES

Floating offshore wind is where Japan's resource actually is — the shelf is narrow and deep water begins close inshore — but no contractor will build the first specialised vessel for a market that does not yet exist. The state broke that deadlock by subsidising the asset rather than the electricity. It is a different instrument from everything else in this report: not a tariff, not a grant to a municipality, but public money placed directly into a private industrial capability. In a Nordic setting that same gap has often been closed by a foundation or a state holding company — which is one reason I want to understand the Danish and Swedish ownership vehicles properly.

UNRESOLVED

One subsidised vessel is not a supply chain. Whether Japanese yards capture floating wind, or whether the work migrates to lower-cost yards abroad while Japan keeps only installation, is undecided.

Ørsted in Japan

A Danish company as the direct bridge — and as the demonstration of a corporate transformation Japan has no equivalent of.

THE COMPANY

Ørsted built the world's first offshore wind farm in Denmark in 1991 and holds roughly **26%** of the global offshore wind market. From 2008 it began shifting from black to green energy: since then it has cut coal consumption by **73%** and halved its carbon emissions. It has **28 offshore wind farms in operation** across Denmark, the UK, Germany, the Netherlands, the US and Taiwan, having installed **1,600 turbines** totalling **7.6 GW**, supplying about **18 million people**, with a further **3.4 GW** under construction and around **3,300** dedicated staff. Roughly **one in four** offshore turbines worldwide was installed by the company. It intends to quadruple its renewable capacity to **50 GW by 2030**.

ITS OWN DECARBONISATION TIMETABLE

2023	Complete exit from coal-fired generation
2025	Carbon neutrality in its own operations
2032	50% of the supply chain decarbonised
2040	Carbon neutrality across the entire business including the supply chain

IN JAPAN

Ørsted entered the Japanese market in 2019 and opened a Tokyo office in May 2019, in order to pursue offshore wind in the Asia-Pacific. Its case for Japan is straightforward: over 70% of the country is mountainous, and about three quarters of 127 million people live in coastal urban areas — so offshore wind is close to the ideal form of large-scale domestic renewable generation.

WHY IT IS IN THIS REPORT

Three reasons. First, it is a literal Denmark–Japan bridge: the questions I want to ask in Copenhagen and Malmö about Danish energy governance have a direct Japanese footprint. Second, it is the counter-example to Case 4 — a single company that took the whole transformation onto its own balance sheet, rather than a consortium assembled per project. Third, and most relevant to our conversation, its history is inseparable from Danish ownership arrangements: a company with a majority state shareholding, in a country with a strong tradition of foundation ownership and cooperative wind, was able to sustain a twenty-year strategic reorientation. No Japanese utility has done anything comparable, and I suspect the reason is structural rather than managerial.

The Ørsted figures above are taken from a Japanese-language corporate profile collected in 2023 and reflect the company's position at that time; they should be refreshed before citation.

Gotō, Nagasaki — E-WIND, citizens' power, and a training association

A small island city that retained locally what a city of 940,000 did not.

THE FIRM

E-WIND Ltd. is based in Tomie-chō, Gotō City, in the Gotō islands off Nagasaki. Founded December 1990; capital ¥15m; **50 staff** at February 2022, average age **39**. It entered wind turbine maintenance in 2006 and consolidated entirely into wind O&M in 2008, when it took its present name. It now runs branch offices in Hokkaido, Muroran, Wakayama, Kagoshima and Fukue, and monitors **more than 100 onshore and offshore turbines** from Hokkaido to Okinawa, 24 hours a day, year-round, from its own information management centre — able to shut a turbine down remotely. Technicians receive about three months of training in the company's own facility followed by a year accompanying senior engineers. The company developed its own tablet-based maintenance management system. It was recognised by METI as a *Regional Future Driving Enterprise* (2017) and among the *300 SMEs taking flight* for regional contribution (2016), and received a special jury prize in the New Energy Awards (2020).

WHAT IT BUILT AROUND ITSELF — THE PART THAT MATTERS

- **Gotō Citizens' Power (2018).** A local retailer, established so that residents and local businesses would experience the benefit of the offshore wind and solar generation around them. E-WIND was closely involved in founding it and supports its product design, marketing and agency network. The retail brand is "Gotō no Denki" — Gotō's electricity.
- **Nagasaki Wind Service Group (2018).** A voluntary association whose explicit purpose is to help *other* firms enter turbine maintenance: a common training curriculum, and visualisation of member technicians' skill levels, to relieve the technician shortage created by the acceleration of the renewables market. A company deliberately creating its own competitors, because the constraint is the labour pool rather than the order book.
- **Its own building.** The head office runs on 100% Gotō-produced renewable energy, with rooftop PV and batteries under an **on-site PPA** and the balance bought from Gotō no Denki. Company EVs double as emergency power and the stored electricity is opened to the community in a disaster. The arrangement is certified under a local "Gotō-version RE100".
- **Farmland regeneration.** Abandoned agricultural land is restored to produce rice, vegetables and camellia oil from Gotō's camellia woods, returned to residents.

WHY I AM PUTTING A 50-PERSON COMPANY BESIDE A 220 MW WIND FARM

Because on the question this report asks, Gotō scores better than Kitakyushu. The turbines around Gotō are owned by outside investors, exactly as at Hibikinada. But the *maintenance* — the twenty-year revenue stream, the skilled employment, the training system and the retail relationship with residents — is owned locally, by a company that then built a citizens' retailer and a training association around itself. Kitakyushu has the port, the consortium and the industrial estate; Gotō has the recurring local income. The Japanese offshore wind industry has, so far, tended to award the construction to Tokyo and the O&M to the region, and Gotō is what it looks like when a region takes that seriously.

UNRESOLVED

Whether the model scales. Gotō's advantage is partly that it is small, remote and cohesive enough for one firm to matter; the same strategy in a city of 940,000 would face incumbents Gotō does not have.

FAIS and the Science and Research Park

Thirty years of environment-and-knowledge-led growth strategy, and a disappointing economic result.

Everything above is what the city presents, and most of it is genuinely working. My own research concerns the longer record, and it is less flattering. Kitakyushu has run an environment-and-knowledge-led growth strategy for three decades through the Science and Research Park and its agency FAIS, and the city's gross product has nonetheless grown slowly relative to other designated cities. The transition has been real as environmental policy and weak as economic policy. I examined this through a growth-contribution analysis of Kitakyushu against Fukuoka City from 1980 to 2020, and through the record of FAIS's own programmes.

THE INDUSTRIES TARGETED WERE NOT THE INDUSTRIES THAT GREW

PERIOD	INDUSTRIES THE CITY AND FAIS SET OUT TO GROW
1994–1998	Imaging and video, aerospace, airport-related industry
1999–2003	Environment (recycling), information technology, welfare, housing, lifestyle and culture, biotechnology
2004–2008	Environment (recycling, eco-premium products, biomass), information technology, welfare devices, biotechnology, next-generation robotics, semiconductors
2019	Semiconductors, car electronics, information and communication (system LSI), robotics

What in fact contributed most to the city's product in the most recent period was health care and social work, real estate, and miscellaneous manufacturing. The target list was rewritten roughly every five years, largely tracking whichever industry national programmes were funding at the time, while the productive structure underneath moved towards lower-value, locally-consumed services.

THREE EXPLANATIONS I AM TESTING

1. **The success criterion was commercialisation, not value added.** FAIS assessed projects on technical feasibility, the prospect of reaching a product, and employment — not on revenue or contribution to the city's economy. A project that reached market counted as a success even if it never grew. National schemes attached to some of the same projects *did* carry revenue targets, and the robotics programme, which used them, is one of the areas that actually expanded.
2. **There was never enough private capital.** No meaningful venture capital layer formed locally. Regional banks fund company formation rather than R&D, so FAIS financed research from its own budget — bounded by a shrinking municipal balance sheet. The comparison I use is Pittsburgh, where university-led renewal was carried by roughly a billion dollars a year of investment, and where venture investors brought a commercial judgement into the decision to fund at all.
3. **The supporting services never agglomerated.** A technology cluster needs accountants, patent and corporate lawyers, marketing and investors within reach. Around the Science and Research Park these are thin, and the city substituted by providing them itself — which keeps the cluster running but caps it at the scale a municipality can fund, and removes the competition that would improve the services.

This case reports work in progress. The contribution analysis is not yet published and the figures behind it are still being revised; I present the structural argument rather than the numbers, and would rather be corrected than believed.

Synthesis

The nine cases compared on the three questions, five findings, and what is structurally missing.

4 The nine cases compared

CASE	WHO OWNS THE ASSET	WHO SUPPLIES THE CAPITAL	WHO CARRIES THE RISK — AND WHAT THE MUNICIPALITY ENDS UP HOLDING
1 Municipal estate PPA	Equipment company	Private, via consortium	Provider carries technical risk; city carries a long tariff obligation. City holds <i>contracts</i> .
2 Reuse-panel demonstration	Leasing company (Tokyo Century)	Private finance	Lessor carries residual-value risk on second-life assets — the first case here where a financial institution takes it. City holds a <i>method</i> .
3 Eco-Town recyclers	Private firms (Mitsubishi Chemical, Ricoh groups)	Corporate parents; national Eco-Town designation	Firms carry commodity-price risk. City holds <i>land, designation and an industry</i> .
4 Hibiki offshore wind	SPC owned by five national companies	Corporate equity and project finance	Consortium carries all of it. City holds <i>tax, a tender record and a port</i> — no equity.
5 Green Energy Port	Public port; private facilities on it	National and municipal port investment	Public sector carries demand risk on the port; firms carry their own. City holds <i>infrastructure and locational advantage</i> .
6 FLOAT RAISER	Private contractors (Toda, Yoshida-gumi)	Ministry of the Environment subsidy plus private	State bought down first-mover risk directly. Nation holds a <i>capability</i> .
7 Ørsted	The company itself	Corporate balance sheet	One firm carries a twenty-year reorientation — possible under Danish ownership arrangements.
8 Gotō / E-WIND	Generation: outside investors. O&M and retail: local	Small local firm; community retailer	Local firm carries operating risk and keeps the recurring income. Region holds <i>skills, jobs and a customer relationship</i> .
9 FAIS / Science Park	Public agency	Municipal budget and national grants	City carries everything, at a scale its budget allows. City holds <i>projects that reached market but did not grow</i> .

5 Five findings

5.1 Kitakyushu buys services; it does not accumulate assets

Read down the ownership column and only Case 5 leaves the municipality owning anything. This is not incompetence — it is the rational response to weak fiscal autonomy and no municipal utility. But it means the city's capability is contractual. When a contract ends, or a provider leaves, or the national grant regime changes, there is little accumulated capital to fall back on. A Nordic city that spent thirty years owning heat networks, housing and water has, by construction, a different set of options in 2026.

5.2 The binding constraint is the absence of a private capital layer

Case 9 says it directly: no venture capital, regional banks that finance company formation rather than R&D, and an agency forced to fund research from a municipal budget. Case 6 shows the state stepping in where private capital would not. Case 2 is the only place where a financial institution voluntarily takes a novel risk — and it is a 5 kW demonstration. Every large project in this report was financed by national corporations, national subsidy, or the city itself. Nothing was financed by a local investor.

5.3 Aggregation is the one instrument that clearly works

Bundling eighteen municipalities and 3,600 buildings turned an unbankable municipal estate into a portfolio private counterparties would bid for. The same logic explains why the port strategy worked and the science park did not: the port aggregated demand for a whole region and a whole industry, while FAIS supported projects one at a time. Where Kitakyushu has created scale it has attracted capital; where it has acted retail, it has had to pay for everything itself.

5.4 The value that stays local is operations, not construction

Construction contracts go to national firms — Penta-Ocean, Nippon Steel Engineering, Vestas, J-POWER. What can stay is the twenty-year maintenance relationship, and Gotō shows a region capturing exactly that, together with the training system that makes it defensible. Kitakyushu has begun to do the same through Hokutaku's shareholding and the O&M port, but Gotō went further by building a citizens' retailer and an association that trains its own competitors.

5.5 Resilience, not carbon, is what unlocks Japanese budgets

The battery at Hiagari Citizens' Centre is justified by evacuation-shelter duty; E-WIND's EVs are justified by disaster power supply; the Kitakyushu energy park literature leads with the city's distance from plate boundaries. Carbon is real but it is rarely the argument that carries a committee. Any transfer of Japanese practice to Europe, or European practice to Japan, has to survive that translation.

6 The missing layers

Setting the nine cases side by side, four things are absent from the Japanese municipal transition that a Nordic reader would expect to find:

A municipal utility	No ownership of generation, networks or heat; a minority position in a retailer is the whole of it. The city cannot cross-subsidise, cannot borrow against an asset base, and cannot use tariffs as policy.
Patient equity	No foundation, holding company or public investment vehicle able to hold an asset for twenty years on a non-commercial mandate. Case 6 shows the state substituting with a one-off subsidy.
A local investor base	No venture capital; regional banks lending against company formation rather than technology. Case 2's leasing company is the first sign of one.
Business services agglomeration	The accountants, patent lawyers and marketing capability a cluster needs are thin, and the city has substituted for them itself — capping scale and removing competition.

Those four absences are, I think, one absence described four ways: **nobody in the region owns capital with a long horizon and a public purpose**. Denmark answers that with foundation ownership and municipal holding companies; Sweden answers it differently again. That is the substance of what I want to learn.

Fourteen questions for the City of Malmö

Grouped by theme. I have deliberately made no claims about Malmö's own figures — I would rather ask than assume.

A. The shipyard and the aftermath

1. Who absorbed the cost?

When Kockums wound down, who carried it — the municipal balance sheet, the state, or private developers? Kitakyushu never had a single closure event; its steel industry shrank slowly and is still there, which may have made the transition politically easier and economically slower.

2. What happened to the workforce?

Japanese transitions are often cushioned by redeployment inside the same corporate group, which protects individuals but slows structural change. Did Sweden's separation of employment protection from the individual firm produce a faster or a more painful adjustment — and which would you choose again?

3. Was Västra Hamnen an economic project or a symbolic one?

Kitakyushu's equivalents — Eco-Town, the Science Park — succeeded and failed respectively, and the difference seems to be whether incumbent industry had a reason to be there. What made the difference in Malmö?

B. Ownership and money

4. What does the City of Malmö still own?

Across district heating, water, housing and any energy assets — and how much of your climate programme depends on that ownership rather than on regulation or procurement?

5. Municipal capital, regulation, or purchasing power?

Roughly what share of your targets is delivered by each? I would like to compare the mix rather than the targets.

6. Is there patient capital with a public purpose?

I am also visiting Lolland and its holding company LOKE, and studying Danish foundation-owned enterprises. Does Malmö have any vehicle that lets public purposes be pursued with private capital on a long horizon — and if not, what fills the gap?

7. Did you ever have to buy down first-mover risk directly?

Case 6 in this report is a construction vessel the Japanese state paid for because no contractor would build the first one. Has Malmö or Region Skåne ever had to fund an asset rather than an outcome?

C. Mechanisms I would like to test against yours

8. Third-party ownership on municipal buildings.

Do you use PPA or energy-service models, and did they survive procurement law? In Japan the weak point is that a long service contract sits badly with annual budgeting; I would expect EU procurement rules to create a different but comparable friction.

9. The price rule.

Kitakyushu will only do a PPA where the tariff is at or below the price it already pays — decarbonisation is not permitted to cost more. Does Malmö operate any equivalent discipline, and has it helped or constrained you?

10. Scale through your neighbours.

Kitakyushu bundled eighteen municipalities and 3,600 buildings to make its programme large enough to attract bidders. How does Malmö achieve scale with Lund, Burlöv and Region Skåne — and does joint procurement work in practice or mainly on paper?

11. Second-life assets.

Case 2 is an attempt to make used solar panels financeable. Sweden's first large wave of renewables will decommission sooner than Japan's. Is anyone building a market in second-life equipment, and will a lessor touch it?

D. The emissions you do not control

12. The 64% problem.

Two-thirds of Kitakyushu's emissions belong to private industry the city cannot regulate and cannot afford to lose. Does Malmö face the same asymmetry — and if so, what has actually worked: agreements, planning conditions, procurement, or public pressure?

13. Keeping the operations work local.

In Japanese offshore wind, the construction goes to national firms and only maintenance can stay in the region. Gotō built a local O&M company, a citizens' retailer and a training association around that fact. Has Malmö deliberately targeted the recurring, operational share of any industry — and did it hold?

E. The question underneath all of it

14. Does owning beat buying?

Kitakyushu buys services and ends thirty years with contracts. Malmö, Lolland and the Danish foundation-owned firms kept ownership. Has that left you with more capability — or only with more debt and slower decisions? An honest answer, including the parts that did not work, is worth more to me than a successful case study.

Sources and method

What this report rests on, and what it does not.

7 Method note

The primary evidence is a set of documents collected during a study visit to Kitakyushu on **22 September 2023** by the Public Finance Study Group of Kyoto University — city presentations, company profiles, project brochures and site materials, several of them prepared for that visit. These are promotional and official documents: they are reliable on structure, dates, capacities and ownership, and they are advocacy on outcomes. I have used them for the former and treated the latter sceptically.

Figures are reported as the source gives them, with the source's date attached, and have not been updated to 2026 — several will have moved, particularly the offshore wind farm's status and Ørsted's portfolio. Case 9 draws on my own unpublished research and is flagged as such in the text. Historical background on the pollution period and the 1963 merger comes from the standard literature on Kitakyushu, not from these documents. **No claims are made about Malmö, Lolland or Swedish and Danish practice**; where I would otherwise have asserted something, Part IV asks it instead.

8 Sources

1. City of Kitakyushu, Environment Bureau, Renewable Energy Promotion Division, 「脱炭素先行地域にかかる北九州市の取り組み」 (*Kitakyushu's initiatives concerning the Decarbonisation Leading Area*), presentation, 22 September 2023. — Cases 1 and 3; Part I.
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8. Ørsted Japan K.K., *Company Profile* (Japanese edition), c. 2022–23. — Case 7.
9. E-WIND Ltd., *Corporate Information*, c. 2022. — Case 8.
10. Author's own research on Kitakyushu's industrial structure and the Science and Research Park (FAIS), drawing on *Kitakyushu City Economic Accounts* (1996–2020) and FAIS programme documents; with Pittsburgh as the comparator. Unpublished. — Case 9.

9 A note on what is missing

Three gaps a reader should hold against this report. It is written almost entirely from Japanese sources and therefore reproduces a Japanese framing of what counts as success. It says nothing about the residents of Kitakyushu, who appear

here only as evacuees sheltering beside a battery — the citizens' movement of the 1960s is the last point at which they act. And it treats the 64% of emissions that belong to heavy industry as a constraint rather than a subject, because the documents I have do so. If our conversation has time for only one thing beyond the questions in Part IV, I would rather it were the third of those.